Human Review: No sparse Euclidean grid in ℓ_∞^n

Original automatic proof: `solution_packet.pdf`
Checked date: 18 June 2026
Status: Manually checked.
Verdict: The proof appears correct.

Human Comments

The proposed argument appears to give a correct negative answer to Fresen's Problem 4, already for the spaces $X = \ell_\infty^n$. The problem asks whether one can choose bases in all n-dimensional Banach spaces so that sufficiently sparse coefficient vectors are uniformly Euclidean. The proof shows that this is impossible in ℓ_∞^n for any fixed distortion constant C, even if the allowed sparsity tends to infinity arbitrarily slowly.

The translation to matrices is clear. A basis of ℓ_∞^n gives a linear map

$$T : \mathbb{R}^n \longrightarrow \ell_\infty^n$$

with rows $r_i = (r_{ij})_{j=1}^n$. The upper sparse estimate implies that, for every row i and every coordinate set A with $|A| \leq k$,

$$\left(\sum_{j \in A} r_{ij}^2 \right)^{1/2} \leq C.$$

This follows by viewing the i-th row as a functional on the Euclidean coordinate subspace supported on A.

Counting Lemma

The counting lemma is the main combinatorial ingredient and appears correct. It says that $2n$ subsets of the signed coordinate set

$$\Omega_n = \{1, \ldots, n\} \times \{-1, 1\},$$

each of size at most Mh, cannot substantially hit every signed h-set once $h \to \infty$ and $h \leq \sqrt{\log n}$.

For a fixed set $H \subseteq \Omega_n$ with $|H| \leq Mh$, the number of signed h-sets meeting H in at least $q_0 = \lceil \eta h \rceil$ points is bounded by

$$\sum_{q=q_0}^{h} \binom{Mh}{q} \binom{n}{h-q} 2^{h-q}.$$

Dividing by the total number $\binom{n}{h} 2^h$ of signed h-sets gives the fraction hit by this particular H. Using

$$\frac{\binom{n}{h-q}}{\binom{n}{h}} \leq \left(\frac{2h}{n} \right)^q$$

for $h \leq n/2$, this fraction is at most a constant-factor variant of

$$h \, 2^{Mh} \left(\frac{h}{n} \right)^{\eta h}.$$

After multiplying by $2n$, the logarithm is bounded above by

$$\log(2n) + \log h + Mh \log 2 - \eta h \log\left(\frac{n}{h}\right),$$

which tends to $-\infty$. The negative term is essentially $-\eta h \log n$, and it dominates the positive terms because $h \to \infty$ while $h \le \sqrt{\log n}$. Hence the union of the $2n$ families cannot cover all signed h-sets.

This part of the proof is correct, although it would be useful in a polished version to expand the counting estimate slightly more than in the current automatic output.

Application to the Counterexample

The proof then chooses

$$h = \left\lfloor \min\left\{\frac{k}{64C^2}, \sqrt{\log n}\right\}\right\rfloor, \qquad \theta = \frac{1}{4\sqrt{h}}.$$

Since $k \to \infty$ and $n \to \infty$, one has $h \to \infty$ directly; there is no real need to pass to a subsequence, apart from discarding finitely many small values if desired.

For each row i, set

$$A_i = \{j : |r_{ij}| \ge \theta\}.$$

The proof first shows that $|A_i| < k$, because otherwise one could choose k elements of A_i and contradict the row mass estimate. This step is needed because the row mass estimate is only known for coordinate sets of size at most k. Once $|A_i| < k$ is known, the same estimate applied to A_i gives

$$|A_i| \le 16C^2 h.$$

For each row i and sign $s \in \{-1, 1\}$, define

$$H_i^s = \{(j, \varepsilon) \in \Omega_n : s\varepsilon r_{ij} \ge \theta\}.$$

Then $|H_i^s| \le 16C^2 h$, so these are precisely the small signed patterns to which the counting lemma will be applied.

Given an arbitrary signed h-set

$$E(S, \varepsilon) = \{(j, \varepsilon_j) : j \in S\},$$

consider the vector $a_j = \varepsilon_j/\sqrt{h}$ on S and $a_j = 0$ outside S. The lower estimate gives a row i and a sign $s \in \{-1, 1\}$ such that

$$\sum_{j \in S} s r_{ij} \varepsilon_j \ge \sqrt{h}.$$

Now define

$$B = \{j \in S : s r_{ij} \varepsilon_j \ge \theta\}.$$

It is helpful to record explicitly that

$$E(S, \varepsilon) \cap H_i^s = \{(j, \varepsilon_j) : j \in B\}.$$

Thus showing that $|B|$ is large is exactly the same as showing that the signed h-set $E(S, \varepsilon)$ has large intersection with one of the sets H_i^s.

The terms outside B contribute at most $h\theta = \sqrt{h}/4$, so the contribution from B is at least $3\sqrt{h}/4$. By Cauchy-Schwarz and the row mass estimate on B,

$$\frac{3\sqrt{h}}{4} \le C|B|^{1/2}.$$

Hence

$$|B| \ge \frac{9}{16C^2}h.$$

Therefore every signed h-set has intersection at least ηh, with $\eta = 9/(16C^2)$, with one of the $2n$ sets H_i^s, each of size at most Mh, with $M = 16C^2$. This contradicts the counting lemma.

Literature Check

A preliminary citation-oriented search found no indexed citing papers for the published Studia Mathematica version of Fresen's article. OpenAlex, Crossref, and OpenCitations all report no citing works for the journal version, and targeted searches for the distinctive Problem 4 terminology did not reveal an existing answer. This does not replace author feedback or a final literature check, but it supports treating the packet as a potentially new negative answer.

Review Outcome

Archive this packet as an apparently correct full negative answer to Fresen's Problem 4, pending author feedback. The argument is sound, but a polished version should expand the counting lemma and make the relation

$$E(S,\varepsilon) \cap H_i^s = \{(j,\varepsilon_j) : j \in B\}$$

explicit when the set B is introduced.